Himanshu Singh

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Education

- **Institute of Engineering & Management** — Kolkata, India
  Bachelor of Technology - Electrical Engineering ; CGPA: 9.2/10 — *2023 - 2027*
- **Kendriya Vidyalaya No.-2 Ishapore** — Kolkata, India
  Higher Secondary (2022): 94.6% , Secondary (2020): 91.4% — *2020 - 2022*

Research Interests

• Machine Learning and AI for Computing Systems • Large Language Models and Intelligent Networked Systems • Wireless Networks and Network Optimization • Energy-Efficient and Sustainable Computing • Data Center, Cloud, and GPU Computing Systems • Stochastic Modeling, Simulation, and Performance Analysis

Technical Skills

- **Languages**: Python, C/C++, SQL, LaTeX
- **Frameworks**: TensorFlow, Keras
- **Tools**: MATLAB/Simulink, PSpice, LTspice, PSIM
- **Platforms**: Windows, Arduino, Raspberry Pi

Research Experience

- **Jadavpur University, Kolkata** — India
  Research Intern: Green Data Center Power Analysis Framework [Project Report] — *May - July 2026*
  - Developed a decoupled bivariate Green Data Center power analysis framework simulating 520 virtual machines across 49 physical server blades (69 CPU cores) for multi-tenant cloud infrastructures.
  - Modeled network traffic using a Non-Homogeneous Poisson Process (NHPP) and fitted backend workloads with Lognormal distributions using Maximum Likelihood Estimation (MLE) and Akaike Information Criterion (AIC) on the Materna GWA Trace-1 dataset.
  - Implemented a linear power superposition model that demonstrated a persistent 6 kW infrastructure power floor despite over 70% reductions in user traffic.
  - **Advisor:** Dr. Bhimsen Tudu, Associate Professor, Department of Power Engineering, Jadavpur University.
- **NIELIT Lucknow** — Lucknow, India
  IoT Intern — *June 2026*
  - **IoT Application Development**: Developed IoT-based solutions using Arduino/NodeMCU and integrated sensors for real-time data acquisition and monitoring.
  - **Embedded Systems & Hardware Interfacing**: Implemented sensor interfacing, actuator control, and embedded programming for connected device applications.

Projects

- **Design & Optimization of a V2G-Enabled Smart City infrastructure using a IEEE 33-Bus Distribution System with Dynamic Energy pricing platform** [Github]
  - Designed an RL environment using 11 state features, 3 control actions (Charge, Discharge, Hold), and 48 half-hour scheduling intervals, improving stability by 25% under peak load.
  - Achieved sub-200 ms control-decision inference latency, demonstrating real-time process-control and automation capability.
  - The integrated BFS-PSO framework demonstrated that coordinated V2G scheduling can meaningfully reduce system power losses. (August '25)
- **Bivariate Modeling and Traffic Profiles for Green Data Center Power Analysis** [Github]
  - Enhanced a decoupled bivariate simulation framework, modeling network traffic and internal compute load as independent stochastic signals.
  - Evaluated cluster-level power consumption using a linear power superposition model, revealing a persistent 6 kW off-peak power floor, 7.4% baseline compute utilization, and a peak demand of 7.5 kW, highlighting energy proportionality challenges in modern cloud data centers. (April '26)
- **AI Data Center Dynamic Load Modelling & Load Frequency Control (LFC) for Grid Stability** | MATLAB/Simulink | Power System Dynamics | Smart Grid Research [Github]
  - Developed a dynamic load model for large-scale AI data centers using stochastic workload modeling and jump-diffusion processes to capture realistic power demand variations.
  - Designing a transient-state AI data center power model incorporating compute, cooling, networking, storage, UPS, and auxiliary loads with energy-proportional characteristics. (June '26 - Ongoing)

Publications

[1] Subhadeep Sarkar, **Himanshu Singh**, Syamantak Doari, Ranjita Chowdhury, Debashis Jana, and Bhimsen Tudu, "A NHPP-Based Stochastic Electrical Power Profiling of LLM Inference Workloads in GPU Clusters," submitted to the *7th International Conference on Electric Power and Renewable Energy (EPREC-2027)*, NIT Jamshedpur, India, 2027. **(Accepted)**

[2] Subhadeep Sarkar, **Himanshu Singh**, Syamantak Doari, Bhimsen Tudu, and Debashis Jana, "A Component-Level Dynamic Load and UPS Reconnection Modelling Framework for Transient Stability Analysis of Grid-Connected Data Centers," submitted to *IEEE Systems Journal*, 2026. **(Accepted)**

Relevant Coursework

- **Mathematics:** Advanced Calculus, Ordinary Differential Equations, Probability & Statistics, Complex Analysis.
- **Core Technical Courses:** Signal & Systems, Control Systems, Electrical Machines I/II, Power Systems I/II, Analog Electronics, EMFT, Microprocessors.
- **Modeling & Optimization:** Queueing Theory, Markov Chains, Stochastic Processes.

Positions of Responsibility

- **Graphics Head, IEEE IAS Student Branch Chapter, IEM Kolkata (2024 - 2025):** Served as the Graphics Head of the IEEE IAS Student Branch Chapter, IEM Kolkata, designing creative content for technical events, workshops, and chapter activities.
- Organized and coordinated various technical workshops and chapter events, contributing to their planning, promotion, and successful execution.

Academic Achievements & Recognitions

- Selected as finalist in a National Level Hackathon (IEEE TechSangam), 11th & 12th August 2025, organized by IEEE Computer Society and IEEE EMBS Pune Chapter.
- Selected as a Delegate (top 2%) in a highly competitive pool of applicants (50000+ applicants) by the committee of Evaluations for the 2025 HPAIR Asia Conference organised by Harvard University - August, 2025.
- Winner in Posternova - Technical Poster presentation competition organised by IEI-IEM Student's Chapter - April, 2025.
- Qualified GATE EE 2026 in 3rd year of Undergraduate.